Lesson plan · for the teacher

Smart school projects: unit check

IoT · Level 4 · Unit check

Time
45 min

Materials
Board, chalk, paper. No devices needed.

Learning goals

- Start from a problem
- Pilot, place and test
- Keep it running and secure
- Data ethics: fair, safe and honest

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which idea will you use first?
- Which question surprised you, and why?
- What will your pupils build or measure first?

Exit ticket

- One idea I will use, and when I will use it

Try it in class tomorrow
Pick the idea from this unit your pupils need most, teach it this week, and tell a colleague how it went.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **Which is the best success measure for the water-tank project?**

- ◯ Three sensors installed and online by the end of March
- ◯ No dry days by the end of term (five last term)
- ◯ Fewer days without water than before

2 **A CO_2 sensor beside an open window usually reads lower than the air most pupils breathe.**

- ◯ True
- ◯ False

3 **Match each practice to the problem it prevents.**

A measurable question	• • • •	People surprised by what's collected
A pilot node	• • • •	A dead sensor that still looks fine
A heartbeat	• • • •	The same mistake built ten times
Asking families first	• • • •	A project nobody can judge

4 **For the tank, a miss means a day without water, and a false alarm means a wasted climb up the tower. Where should the low-level alert sit?**

- ◯ Early: accept some false alarms so real low levels aren't missed
- ◯ It doesn't matter much: both mistakes cost about the same
- ◯ Very low: alert only when nearly empty, so there are no false alarms

5 **Two AA batteries are rated 2,500 mAh, and the node averages 1.75 mA. Plan with 70% of the label.**

Usable capacity: _____ mAh. Battery life: about _____ hours, roughly six weeks.

Word bank: 1,750 · 1,000 · 2,500 · 1,429